

Oracle DBA Cheat Sheet

Topic	Details	Check Query/Command
Oracle Database	Robust RDBMS for high-performance data management.	SELECT * FROM v\$version;
RMAN Backup	Consistent (Cold, offline, MOUNT mode), Inconsistent (Hot, online, ARCHIVELOG mode), Full (all data), Incremental (daily changes).	RMAN> report schema; RMAN> list backup;
RMAN Cold Backup	SHUTDOWN IMMEDIATE; STARTUP MOUNT; BACKUP DATABASE;	SELECT status FROM v\$instance;
RMAN Hot Backup	BACKUP DATABASE PLUS ARCHIVELOG; (Requires ARCHIVELOG mode).	ARCHIVE LOG LIST;
Data Guard	Standby: Clone of production DB. Types: Physical (block-by-block, same DB ID), Logical (SQL apply, separate DB ID), Snapshot (testing with flashback). Processes: LNS (redo transport), MRP (physical apply), LSP (logical apply). Modes: Max Performance, Max Availability, Max Protection.	SELECT protection_mode FROM v\$database; SELECT database_role FROM v\$database; SELECT switchover_status FROM v\$database; SELECT process, status FROM v\$managed_standby;
Data Guard Parameters	DB_NAME, DB_UNIQUE_NAME, LOG_ARCHIVE_DEST_n, FAL_SERVER, STANDBY_FILE_MANAGEMENT, LOG_ARCHIVE_CONFIG.	SHOW PARAMETER db_name; SHOW PARAMETER db_unique_name; SHOW PARAMETER log_archive_dest; SHOW PARAMETER fal_server; SHOW PARAMETER standby_file_management; SHOW PARAMETER log_archive_config;
RAC	One database, multiple instances for scalability.	SELECT inst_id, instance_name, status FROM gv\$instance;
ASM	Automatic Storage Management for simplified storage.	SELECT name, state FROM v\$asm_diskgroup;
GoldenGate	Real-time data replication using capture/apply.	SELECT capture_name, status FROM dba_capture; SELECT apply_name, status FROM dba_apply;
Oracle Architecture	Instance: SGA (Database Buffer Cache: stores data blocks; Redo Log Buffer: holds redo entries; Shared Pool: SQL, PL/SQL, data dictionary, library cache; Large Pool: RMAN, parallel execution; Java Pool: Java execution), PGA (session memory for sorting), Background Processes (DBWn: writes buffers; LGWR: writes redo logs; CKPT: updates headers; SMON: recovery; PMON: cleans failed processes). Database: Datafiles, control files, redo logs.	SELECT name, value FROM v\$sga; SELECT component, current_size/1024/1024 MB FROM v\$sga_dynamic_components; SELECT process FROM v\$bgprocess WHERE paddr != '00'; SELECT name, value/1024/1024 MB FROM v\$pgastat WHERE name IN ('aggregate PGA target parameter', 'total PGA allocated');

User Process	Client process initiating SQL requests.	<code>SELECT sid, username, program FROM v\$session WHERE type='USER';</code>
Server Process	Executes SQL for user.	<code>SELECT sid, spid, program FROM v\$process WHERE addr IN (SELECT paddr FROM v\$session);</code>
Physical Files	Datafiles (.dbf), control files (.ctl), redo logs (.log), archive logs, spfile/pfile.	<code>SELECT name FROM v\$datafile; SELECT name FROM v\$controlfile; SELECT member FROM v\$logfile;</code>
Logical Files	Tablespaces (8KB default block size), segments, extents, blocks.	<code>SELECT tablespace_name, segment_name FROM dba_segments;</code>
File Paths	Datafiles: /u01/oradata/DBNAME/, Control: /u01/control/, Redo: /u01/redolog/.	<code>SELECT file_name FROM dba_data_files; SELECT name FROM v\$controlfile; SELECT member FROM v\$logfile;</code>
TNS	Transparent Network Substrate (TNSNAMES.ORA).	<code>tnsping <service_name></code>
DB Link	Access remote database objects.	<code>SELECT db_link, host FROM dba_db_links;</code>
Patching	Resolves bugs and vulnerabilities.	<code>opatch lsinventory;</code>
Hot Backup	<code>ALTER TABLESPACE tbs BEGIN BACKUP; copy files; ALTER TABLESPACE tbs END BACKUP;.</code>	<code>SELECT tablespace_name, status FROM dba_tablespaces WHERE status='BACKUP';</code>
EXPDP (Full Backup)	<code>expdp system/pwd FULL=Y DIRECTORY=dir DUMPFILE=full_exp.dmp LOGFILE=full_exp.log</code>	<code>SELECT job_name, operation, state FROM dba_datapump_jobs WHERE operation='EXPORT';</code>
EXPDP (Schema/Table)	<code>expdp system/pwd SCHEMAS=schema_name DIRECTORY=dir DUMPFILE=schema.dmp LOGFILE=schema.log</code>	<code>SELECT job_name, operation, state FROM dba_datapump_jobs WHERE operation='EXPORT';</code>
IMPDP	<code>impdp system/pwd DIRECTORY=dir DUMPFILE=full_exp.dmp LOGFILE=full_imp.log</code>	<code>SELECT job_name, operation, state FROM dba_datapump_jobs WHERE operation='IMPORT';</code>
Data Pump Tuning	Use <code>DIRECT=Y, PARALLEL=n</code> , schedule via crontab.	<code>SELECT job_name, degree FROM dba_datapump_jobs;</code>
Data Pump Directory	OS: <code>mkdir /path/dir</code> , DB: <code>CREATE DIRECTORY dir AS '/path/dir';</code>	<code>SELECT directory_name, directory_path FROM dba_directories;</code>
Network Import	<code>impdp system/pwd NETWORK_LINK=dblink DUMPFILE=exp.dmp LOGFILE=imp.log</code>	<code>SELECT db_link, host FROM dba_db_links;</code>
Grant Privilege	<code>GRANT SELECT ON table TO user;</code>	<code>SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE grantee='USER';</code>
Revoke Privilege	<code>REVOKE SELECT ON table FROM user;</code>	<code>SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE grantee='USER';</code>

User Management	CREATE USER user IDENTIFIED BY pwd; ALTER USER user IDENTIFIED BY new_pwd; DROP USER user;	SELECT username, account_status FROM dba_users;
Role Management	CREATE ROLE role_name; GRANT SELECT ON table TO role_name;	SELECT role, privilege FROM dba_role_privs WHERE grantee='ROLE_NAME';
Profile Management	CREATE PROFILE prof LIMIT PASSWORD_LIFE_TIME 180;	SELECT profile, resource_name, limit FROM dba_profiles WHERE profile='PROF';
Add Datafile	ALTER TABLESPACE tbs ADD DATAFILE '/path/file.dbf' SIZE 100M;	SELECT file_name, bytes/1024/1024 MB FROM dba_data_files WHERE tablespace_name='TBS';
Resize Datafile	ALTER DATABASE DATAFILE '/path/file.dbf' RESIZE 200M;	SELECT file_name, bytes/1024/1024 MB FROM dba_data_files WHERE file_name='/path/file.dbf';
Check Tablespace Size	SELECT tablespace_name, SUM(bytes)/1024/1024 MB FROM dba_data_files GROUP BY tablespace_name;	SELECT tablespace_name, SUM(bytes)/1024/1024 MB FROM dba_data_files GROUP BY tablespace_name;
Alter Tablespace	ALTER TABLESPACE tbs ADD DATAFILE '/path/file.dbf' SIZE 100M AUTOEXTEND ON;	SELECT tablespace_name, file_name, autoextensible FROM dba_data_files WHERE tablespace_name='TBS';
SQL Utility	Clear SQL terminal: cl scr	cl scr
DBA Handover Checks	Check alert log, backups, tablespace space, listener, RAC health, Data Guard sync, job schedules.	SELECT * FROM v\$alert_log; SELECT destination, status FROM v\$archive_dest; SELECT tablespace_name, (SUM(bytes)/1024/1024) - (SUM(free_space)/1024/1024) MB_USED FROM dba_data_files JOIN dba_free_space USING(tablespace_name) GROUP BY tablespace_name; lsnrctl status; SELECT inst_id, status FROM gv\$instance; SELECT sequence#, applied FROM v\$archived_log; SELECT job_name, state FROM dba_scheduler_jobs;